TABLE 1 OVERALL Before

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
Total	Before	1471743	353	328	22.2865

TABLE 1 OVERALL During

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
Total	During	1001387	294	275	27.4619

TABLE 1 Before: age_hivdxrange

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	Before	46	0	0	0.0000
18-34	Before	199565	8	8	4.0087
35-49	Before	403457	34	32	7.9315
50-64	Before	639788	143	137	21.4133
65 and over	Before	228887	168	151	65.9714

TABLE 1 During: age_hivdxrange

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	During	27	0	0	0.0000
18-34	During	137269	11	11	8.0135
35-49	During	276370	39	36	13.0260
50-64	During	435203	135	128	29.4116
65 and over	During	152518	109	100	65.5660

TABLE 1 Before: current_gender2

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	Before	.	0	0	.
F	Before	388162	104	101	26.0201
M	Before	1083581	249	227	20.9491

TABLE 1 During: current_gender2

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	During	.	0	0	.
F	During	263575	91	89	33.7665
M	During	737812	203	186	25.2097

TABLE 1 Before: raceeth

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	Before	.	0	0	.
Hispanic	Before	381696	90	81	21.2211
NHB	Before	644269	190	182	28.2491
NHW	Before	413939	67	59	14.2533
Other	Before	31839	6	6	18.8448

TABLE 1 During: raceeth

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	During	.	0	0	.
Hispanic	During	261531	47	44	16.8240
NHB	During	437001	174	165	37.7574
NHW	During	281200	69	62	22.0484
Other	During	21655	4	4	18.4715

TABLE 1 Before: USborn

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	Before	.	0	0	.
No	Before	401837	81	78	19.4109
Unknown	Before	39632	3	3	7.5696
Yes	Before	1030274	269	247	23.9742

TABLE 1 During: USborn

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	During	.	0	0	.
No	During	275347	63	62	22.5170
Unknown	During	27123	3	3	11.0607
Yes	During	698917	228	210	30.0465

TABLE 1 Before: rucaccat2010new

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	Before	.	0	0	.
rural	Before	38746	9	9	23.2282
urban	Before	1432997	344	319	22.2610

TABLE 1 During: rucaccat2010new

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	During	.	0	0	.
rural	During	26301	4	4	15.2085
urban	During	975086	290	271	27.7924

TABLE 1 Before: categ

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	Before	.	0	0	.
IDU	Before	141323	49	49	34.6723
MSM	Before	734777	111	111	15.1066
hetero	Before	492205	151	151	30.6783
other	Before	103438	42	17	16.4350

TABLE 1 During: categ

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	During	.	0	0	.
IDU	During	94668	35	35	36.9713
MSM	During	502508	94	94	18.7062
hetero	During	333735	134	134	40.1516
other	During	70476	31	12	17.0271

TABLE 1 Before: SuppressedVL2020

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
.	Before	.	0	0	.
0	Before	477593	115	107	22.4040
1	Before	994150	238	221	22.2300

TABLE 1 During: SuppressedVL2021

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
.	During	.	0	0	.
0	During	312495	129	119	38.0806
1	During	688892	165	156	22.6451

TABLE 1 Before: theme

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	Before	.	0	0	.
high	Before	998703	265	250	25.0325
low	Before	144823	26	22	15.1910
medium	Before	328217	62	56	17.0619

TABLE 1 During: theme

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	During	.	0	0	.
high	During	678690	223	208	30.6473
low	During	98848	19	17	17.1981
medium	During	223849	52	50	22.3365

TABLE 1 Before: theme1

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	Before	.	0	0	.
high	Before	831845	226	214	25.7259
low	Before	215535	31	29	13.4549
medium	Before	424363	96	85	20.0300

TABLE 1 During: theme1

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	During	.	0	0	.
high	During	564773	197	182	32.2253
low	During	147280	24	22	14.9375
medium	During	289334	73	71	24.5391

TABLE 1 Before: theme2

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	Before	.	0	0	.
high	Before	662890	187	174	26.2487
low	Before	315820	58	53	16.7817
medium	Before	493033	108	101	20.4854

TABLE 1 During: theme2

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	During	.	0	0	.
high	During	449875	153	143	31.7866
low	During	215647	52	50	23.1860
medium	During	335865	89	82	24.4146

TABLE 1 Before: theme3

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	Before	.	0	0	.
high	Before	891143	234	216	24.2385
low	Before	163310	36	32	19.5946
medium	Before	417290	83	80	19.1713

TABLE 1 During: theme3

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	During	.	0	0	.
high	During	606883	197	182	29.9893
low	During	110901	30	28	25.2477
medium	During	283603	67	65	22.9194

TABLE 1 Before: theme4

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	Before	.	0	0	.
high	Before	883186	234	222	25.1363
low	Before	130486	31	24	18.3928
medium	Before	458071	88	82	17.9012

TABLE 1 During: theme4

level	period	wtd_person_time	raw_covid_deaths	wtd_covid_deaths	rate_per_100k
	During	.	0	0	.
high	During	600254	194	182	30.3205
low	During	88927	21	17	19.1168
medium	During	312206	79	76	24.3429

TABLE 2: Before vaccine â€" Model 1 (individual factors)

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119316

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
SuppressedVL2020	0	1		
	1	0		

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
119316	350	6743	112223

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.664	7263.553
AIC	7545.664	7287.553
SBC	7545.664	7333.848

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	285.4392	12	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	195.1094	<.0001
current_gender2	1	1.2397	0.2655
raceeth	3	23.9813	<.0001
USborn	1	12.2147	0.0005
categ	3	11.7144	0.0084
SuppressedVL2020	1	0.3380	0.5610

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.63859	0.39570	2.6045	0.1066	1.894	0.872	4.113	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.70145	0.36141	22.1633	<.0001	5.482	2.700	11.132	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.82979	0.36252	60.9331	<.0001	16.942	8.325	34.478	age_hivdxrange 65 and over
current_gender2	F	1	-0.15484	0.13907	1.2397	0.2655	0.857	0.652	1.125	current_gender2 F
raceeth	Hispanic	1	0.83568	0.18633	20.1155	<.0001	2.306	1.601	3.323	raceeth Hispanic
raceeth	NHB	1	0.78676	0.17466	20.2913	<.0001	2.196	1.560	3.093	raceeth NHB
raceeth	Other	1	0.58530	0.43136	1.8411	0.1748	1.796	0.771	4.182	raceeth Other
USborn	No	1	-0.48861	0.13980	12.2147	0.0005	0.613	0.466	0.807	USborn No
categ	IDU	1	0.35132	0.18780	3.4995	0.0614	1.421	0.983	2.053	categ IDU
categ	hetero	1	0.41209	0.16575	6.1813	0.0129	1.510	1.091	2.090	categ hetero
categ	other	1	-0.22839	0.28870	0.6259	0.4289	0.796	0.452	1.401	categ other
SuppressedVL2020	0	1	0.07116	0.12240	0.3380	0.5610	1.074	0.845	1.365	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2020 specified through March 31 of the following year 0

TABLE 2: During vaccine â€" Model 1 (individual factors)

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116787

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
SuppressedVL2021	0	1		
	1	0		

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
116787	291	4307	112189

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.844	6125.831
AIC	6306.844	6149.831
SBC	6306.844	6193.910

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	206.2297	12	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	103.9464	<.0001
current_gender2	1	0.5351	0.4645
raceeth	3	11.6707	0.0086
USborn	1	6.6245	0.0101
categ	3	13.9003	0.0030
SuppressedVL2021	1	21.1677	<.0001

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.56368	0.35765	2.4839	0.1150	1.757	0.872	3.542	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.42658	0.32486	19.2846	<.0001	4.164	2.203	7.872	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.19981	0.32965	44.5307	<.0001	9.023	4.729	17.217	age_hivdxrange 65 and over
current_gender2	F	1	-0.11530	0.15762	0.5351	0.4645	0.891	0.654	1.214	current_gender2 F
raceeth	Hispanic	1	0.04065	0.21520	0.0357	0.8502	1.041	0.683	1.588	raceeth Hispanic
raceeth	NHB	1	0.51982	0.17720	8.6057	0.0034	1.682	1.188	2.380	raceeth NHB
raceeth	Other	1	0.01662	0.51675	0.0010	0.9743	1.017	0.369	2.799	raceeth Other
USborn	No	1	-0.40930	0.15903	6.6245	0.0101	0.664	0.486	0.907	USborn No
categ	IDU	1	0.22259	0.20831	1.1418	0.2853	1.249	0.831	1.879	categ IDU
categ	hetero	1	0.44123	0.18511	5.6815	0.0171	1.555	1.082	2.235	categ hetero
categ	other	1	-0.51039	0.33695	2.2945	0.1298	0.600	0.310	1.162	categ other
SuppressedVL2021	0	1	0.57987	0.12604	21.1677	<.0001	1.786	1.395	2.286	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2021 specified through March 31 of the following year 0

TABLE 2: Before vaccine â€" Final Model (individual + community)

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119316

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2020	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
119316	350	6743	112223

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.664	7258.815
AIC	7545.664	7290.815
SBC	7545.664	7352.542

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	291.6293	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	193.8708	<.0001
current_gender2	1	1.3702	0.2418
raceeth	3	20.8606	0.0001
USborn	1	12.2185	0.0005
rucaccat2010new	1	0.1344	0.7139
categ	3	11.3736	0.0099
SuppressedVL2020	1	0.2768	0.5988
theme	2	4.1185	0.1275
percent_vacc	1	0.0548	0.8149

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.63413	0.39574	2.5677	0.1091	1.885	0.868	4.095	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.69229	0.36143	21.9234	<.0001	5.432	2.675	11.031	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.81990	0.36245	60.5312	<.0001	16.775	8.244	34.134	age_hivdxrange 65 and over
current_gender2	F	1	-0.16225	0.13861	1.3702	0.2418	0.850	0.648	1.116	current_gender2 F
raceeth	Hispanic	1	0.80422	0.18724	18.4475	<.0001	2.235	1.548	3.226	raceeth Hispanic
raceeth	NHB	1	0.72100	0.17726	16.5449	<.0001	2.056	1.453	2.911	raceeth NHB
raceeth	Other	1	0.55434	0.42996	1.6623	0.1973	1.741	0.749	4.043	raceeth Other
USborn	No	1	-0.50686	0.14500	12.2185	0.0005	0.602	0.453	0.800	USborn No
rucaccat2010new	rural	1	-0.12752	0.34780	0.1344	0.7139	0.880	0.445	1.740	rucaccat2010new

										rural
categ	**IDU**	1	0.34950	0.18791	3.4594	0.0629	1.418	0.981	2.050	categ IDU
categ	**hetero**	1	0.40808	0.16617	6.0309	0.0141	1.504	1.086	2.083	categ hetero
categ	**other**	1	-0.21774	0.28970	0.5649	0.4523	0.804	0.456	1.419	categ other
SuppressedVL2020	**0**	1	0.06477	0.12312	0.2768	0.5988	1.067	0.838	1.358	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2020 specified through March 31 of the following year 0
theme	**high**	1	0.39392	0.23988	2.6968	0.1006	1.483	0.927	2.373	theme high
theme	**medium**	1	0.15399	0.25884	0.3539	0.5519	1.166	0.702	1.937	theme medium
percent_vacc		1	-0.12686	0.54192	0.0548	0.8149	0.881	0.305	2.548	Percent of populatin +5 vaccinated

TABLE 2: During vaccine â€" Final Model (individual + community)

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116787

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2021	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
116787	291	4307	112189

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.844	6116.355
AIC	6306.844	6148.355
SBC	6306.844	6207.128

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	215.1058	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	105.0483	<.0001
current_gender2	1	0.6989	0.4031
raceeth	3	9.4521	0.0238
USborn	1	5.4309	0.0198
rucaccat2010new	1	3.2905	0.0697
categ	3	13.5369	0.0036
SuppressedVL2021	1	21.6229	<.0001
theme	2	5.2429	0.0727
percent_vacc	1	1.9057	0.1674

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.56541	0.35769	2.4988	0.1139	1.760	0.873	3.548	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.43135	0.32506	19.3894	<.0001	4.184	2.213	7.913	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.21104	0.32960	45.0021	<.0001	9.125	4.783	17.410	age_hivdxrange 65 and over
current_gender2	F	1	-0.13190	0.15778	0.6989	0.4031	0.876	0.643	1.194	current_gender2 F
raceeth	Hispanic	1	0.03084	0.21472	0.0206	0.8858	1.031	0.677	1.571	raceeth Hispanic
raceeth	NHB	1	0.46846	0.17826	6.9065	0.0086	1.598	1.126	2.266	raceeth NHB
raceeth	Other	1	-0.01549	0.51636	0.0009	0.9761	0.985	0.358	2.709	raceeth Other
USborn	No	1	-0.38551	0.16543	5.4309	0.0198	0.680	0.492	0.941	USborn No
rucaccat2010new	rural	1	-0.93013	0.51276	3.2905	0.0697	0.395	0.144	1.078	rucaccat2010new

										rural
categ	**IDU**	1	0.20670	0.20850	0.9829	0.3215	1.230	0.817	1.850	categ IDU
categ	**hetero**	1	0.43186	0.18501	5.4489	0.0196	1.540	1.072	2.213	categ hetero
categ	**other**	1	-0.50917	0.33845	2.2634	0.1325	0.601	0.310	1.167	categ other
SuppressedVL2021	**0**	1	0.58258	0.12528	21.6229	<.0001	1.791	1.401	2.289	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2021 specified through March 31 of the following year 0
theme	**high**	1	0.56501	0.26898	4.4123	0.0357	1.759	1.039	2.981	theme high
theme	**medium**	1	0.35587	0.29300	1.4752	0.2245	1.427	0.804	2.535	theme medium
percent_vacc		1	-0.76964	0.55752	1.9057	0.1674	0.463	0.155	1.381	Percent of populatin +5 vaccinated

TABLE 3: Before vaccine â€" theme1

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119316

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme1	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2020	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
119316	350	6743	112223

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.664	7258.327
AIC	7545.664	7290.327
SBC	7545.664	7352.054

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	291.2577	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	193.9105	<.0001
current_gender2	1	1.3699	0.2418
raceeth	3	20.5330	0.0001
USborn	1	11.9992	0.0005
rucaccat2010new	1	0.0955	0.7572
categ	3	11.3428	0.0100
SuppressedVL2020	1	0.2849	0.5935
theme1	2	4.5454	0.1030
percent_vacc	1	0.0124	0.9112

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.63632	0.39560	2.5873	0.1077	1.890	0.870	4.103	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.69301	0.36125	21.9643	<.0001	5.436	2.678	11.035	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.82096	0.36225	60.6432	<.0001	16.793	8.256	34.157	age_hivdxrange 65 and over
current_gender2	F	1	-0.16237	0.13872	1.3699	0.2418	0.850	0.648	1.116	current_gender2 F
raceeth	Hispanic	1	0.80509	0.18784	18.3691	<.0001	2.237	1.548	3.232	raceeth Hispanic
raceeth	NHB	1	0.71552	0.17896	15.9862	<.0001	2.045	1.440	2.905	raceeth NHB
raceeth	Other	1	0.55113	0.43006	1.6423	0.2000	1.735	0.747	4.031	raceeth Other
USborn	No	1	-0.50341	0.14533	11.9992	0.0005	0.604	0.455	0.804	USborn No
rucaccat2010new	rural	1	-0.10585	0.34243	0.0955	0.7572	0.900	0.460	1.760	rucaccat2010new

										rural
categ	**IDU**	1	0.34530	0.18786	3.3784	0.0661	1.412	0.977	2.041	categ IDU
categ	**hetero**	1	0.40543	0.16571	5.9858	0.0144	1.500	1.084	2.076	categ hetero
categ	**other**	1	-0.22349	0.28973	0.5950	0.4405	0.800	0.453	1.411	categ other
SuppressedVL2020	**0**	1	0.06580	0.12327	0.2849	0.5935	1.068	0.839	1.360	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2020 specified through March 31 of the following year 0
theme1	**high**	1	0.44178	0.20821	4.5021	0.0339	1.555	1.034	2.339	theme1 high
theme1	**medium**	1	0.35322	0.22041	2.5681	0.1090	1.424	0.924	2.193	theme1 medium
percent_vacc		1	0.05684	0.50977	0.0124	0.9112	1.058	0.390	2.875	Percent of populatin +5 vaccinated

TABLE 3: Before vaccine â€" theme2

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119316

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme2	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2020	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
119316	350	6743	112223

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.664	7259.931
AIC	7545.664	7291.931
SBC	7545.664	7353.658

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	288.3568	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	193.3801	<.0001
current_gender2	1	1.4155	0.2341
raceeth	3	23.2133	<.0001
USborn	1	12.1017	0.0005
rucaccat2010new	1	0.0156	0.9007
categ	3	11.3292	0.0101
SuppressedVL2020	1	0.3430	0.5581
theme2	2	3.3839	0.1842
percent_vacc	1	0.0463	0.8296

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.63838	0.39555	2.6046	0.1066	1.893	0.872	4.111	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.69590	0.36131	22.0312	<.0001	5.452	2.685	11.068	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.81719	0.36232	60.4560	<.0001	16.730	8.224	34.033	age_hivdxrange 65 and over
current_gender2	F	1	-0.16511	0.13878	1.4155	0.2341	0.848	0.646	1.113	current_gender2 F
raceeth	Hispanic	1	0.83513	0.18835	19.6590	<.0001	2.305	1.594	3.334	raceeth Hispanic
raceeth	NHB	1	0.77334	0.17536	19.4486	<.0001	2.167	1.537	3.056	raceeth NHB
raceeth	Other	1	0.57755	0.43024	1.8020	0.1795	1.782	0.767	4.140	raceeth Other
USborn	No	1	-0.50267	0.14450	12.1017	0.0005	0.605	0.456	0.803	USborn No
rucaccat2010new	rural	1	-0.04301	0.34460	0.0156	0.9007	0.958	0.488	1.882	rucaccat2010new

										rural
categ	**IDU**	1	0.34609	0.18772	3.3992	0.0652	1.414	0.978	2.042	categ IDU
categ	**hetero**	1	0.40152	0.16588	5.8587	0.0155	1.494	1.079	2.068	categ hetero
categ	**other**	1	-0.23119	0.28909	0.6395	0.4239	0.794	0.450	1.399	categ other
SuppressedVL2020	**0**	1	0.07213	0.12315	0.3430	0.5581	1.075	0.844	1.368	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2020 specified through March 31 of the following year 0
theme2	**high**	1	0.28110	0.15809	3.1616	0.0754	1.325	0.972	1.806	theme2 high
theme2	**medium**	1	0.15760	0.17260	0.8338	0.3612	1.171	0.835	1.642	theme2 medium
percent_vacc		1	0.11178	0.51928	0.0463	0.8296	1.118	0.404	3.094	Percent of populatin +5 vaccinated

TABLE 3: Before vaccine â€" theme3

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119316

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme3	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2020	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
119316	350	6743	112223

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.664	7261.933
AIC	7545.664	7293.933
SBC	7545.664	7355.660

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	290.1980	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	194.3942	<.0001
current_gender2	1	1.2559	0.2624
raceeth	3	22.0197	<.0001
USborn	1	12.4513	0.0004
rucaccat2010new	1	0.0450	0.8320
categ	3	11.5929	0.0089
SuppressedVL2020	1	0.2979	0.5852
theme3	2	1.4666	0.4803
percent_vacc	1	0.0438	0.8342

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.63758	0.39558	2.5978	0.1070	1.892	0.871	4.108	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.69787	0.36117	22.0992	<.0001	5.462	2.691	11.087	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.82452	0.36212	60.8391	<.0001	16.853	8.288	34.270	age_hivdxrange 65 and over
current_gender2	F	1	-0.15564	0.13888	1.2559	0.2624	0.856	0.652	1.124	current_gender2 F
raceeth	Hispanic	1	0.81749	0.19008	18.4971	<.0001	2.265	1.560	3.287	raceeth Hispanic
raceeth	NHB	1	0.77368	0.17811	18.8686	<.0001	2.168	1.529	3.073	raceeth NHB
raceeth	Other	1	0.58073	0.43055	1.8193	0.1774	1.787	0.769	4.156	raceeth Other
USborn	No	1	-0.51213	0.14514	12.4513	0.0004	0.599	0.451	0.796	USborn No
rucaccat2010new	rural	1	-0.07312	0.34474	0.0450	0.8320	0.929	0.473	1.827	rucaccat2010new

										rural
categ	**IDU**	1	0.35332	0.18781	3.5393	0.0599	1.424	0.985	2.057	categ IDU
categ	**hetero**	1	0.40637	0.16647	5.9592	0.0146	1.501	1.083	2.081	categ hetero
categ	**other**	1	-0.23420	0.28914	0.6561	0.4179	0.791	0.449	1.394	categ other
SuppressedVL2020	**0**	1	0.06734	0.12338	0.2979	0.5852	1.070	0.840	1.362	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2020 specified through March 31 of the following year 0
theme3	**high**	1	0.06805	0.22348	0.0927	0.7608	1.070	0.691	1.659	theme3 high
theme3	**medium**	1	-0.11088	0.21534	0.2651	0.6066	0.895	0.587	1.365	theme3 medium
percent_vacc		1	-0.12387	0.59196	0.0438	0.8342	0.883	0.277	2.819	Percent of populatin +5 vaccinated

TABLE 3: Before vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119316

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme4	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2020	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
119316	350	6743	112223

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.664	7259.279
AIC	7545.664	7291.279
SBC	7545.664	7353.006

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	287.3349	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	194.2649	<.0001
current_gender2	1	1.2641	0.2609
raceeth	3	21.9579	<.0001
USborn	1	11.8841	0.0006
rucaccat2010new	1	0.1803	0.6711
categ	3	11.6750	0.0086
SuppressedVL2020	1	0.2559	0.6130
theme4	2	3.8396	0.1466
percent_vacc	1	0.0095	0.9224

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.63463	0.39555	2.5742	0.1086	1.886	0.869	4.095	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.69196	0.36110	21.9551	<.0001	5.430	2.676	11.020	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.81998	0.36203	60.6752	<.0001	16.777	8.252	34.108	age_hivdxrange 65 and over
current_gender2	F	1	-0.15596	0.13871	1.2641	0.2609	0.856	0.652	1.123	current_gender2 F
raceeth	Hispanic	1	0.82173	0.18771	19.1644	<.0001	2.274	1.574	3.286	raceeth Hispanic
raceeth	NHB	1	0.74093	0.17575	17.7727	<.0001	2.098	1.487	2.961	raceeth NHB
raceeth	Other	1	0.57156	0.43047	1.7629	0.1843	1.771	0.762	4.118	raceeth Other
USborn	No	1	-0.49999	0.14504	11.8841	0.0006	0.607	0.456	0.806	USborn No
rucaccat2010new	rural	1	-0.14906	0.35106	0.1803	0.6711	0.862	0.433	1.714	rucaccat2010new

										rural
categ	**IDU**	1	0.35065	0.18793	3.4814	0.0621	1.420	0.982	2.052	categ IDU
categ	**hetero**	1	0.41627	0.16622	6.2714	0.0123	1.516	1.095	2.100	categ hetero
categ	**other**	1	-0.21788	0.28923	0.5675	0.4513	0.804	0.456	1.418	categ other
SuppressedVL2020	**0**	1	0.06227	0.12309	0.2559	0.6130	1.064	0.836	1.355	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2020 specified through March 31 of the following year 0
theme4	**high**	1	0.23664	0.22535	1.1027	0.2937	1.267	0.815	1.971	theme4 high
theme4	**medium**	1	-0.01520	0.23693	0.0041	0.9488	0.985	0.619	1.567	theme4 medium
percent_vacc		1	-0.05206	0.53459	0.0095	0.9224	0.949	0.333	2.707	Percent of populatin +5 vaccinated

TABLE 3: During vaccine â€" theme1

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116787

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme1	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2021	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
116787	291	4307	112189

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.844	6114.875
AIC	6306.844	6146.875
SBC	6306.844	6205.648

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	213.7363	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	104.9669	<.0001
current_gender2	1	0.7051	0.4011
raceeth	3	9.1336	0.0276
USborn	1	5.3255	0.0210
rucaccat2010new	1	3.2679	0.0706
categ	3	13.4669	0.0037
SuppressedVL2021	1	21.6593	<.0001
theme1	2	6.2389	0.0442
percent_vacc	1	1.1828	0.2768

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.56808	0.35776	2.5213	0.1123	1.765	0.875	3.558	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.43099	0.32505	19.3809	<.0001	4.183	2.212	7.910	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.21242	0.32969	45.0326	<.0001	9.138	4.789	17.437	age_hivdxrange 65 and over
current_gender2	F	1	-0.13257	0.15787	0.7051	0.4011	0.876	0.643	1.193	current_gender2 F
raceeth	Hispanic	1	0.03437	0.21451	0.0257	0.8727	1.035	0.680	1.576	raceeth Hispanic
raceeth	NHB	1	0.46260	0.17946	6.6446	0.0099	1.588	1.117	2.258	raceeth NHB
raceeth	Other	1	-0.01604	0.51702	0.0010	0.9752	0.984	0.357	2.711	raceeth Other
USborn	No	1	-0.38283	0.16589	5.3255	0.0210	0.682	0.493	0.944	USborn No
rucaccat2010new	rural	1	-0.91302	0.50506	3.2679	0.0706	0.401	0.149	1.080	rucaccat2010new

										rural
categ	**IDU**	1	0.20145	0.20808	0.9373	0.3330	1.223	0.814	1.839	categ IDU
categ	**hetero**	1	0.42786	0.18476	5.3628	0.0206	1.534	1.068	2.203	categ hetero
categ	**other**	1	-0.51506	0.33800	2.3222	0.1275	0.597	0.308	1.159	categ other
SuppressedVL2021	**0**	1	0.58376	0.12543	21.6593	<.0001	1.793	1.402	2.292	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2021 specified through March 31 of the following year 0
theme1	**high**	1	0.58707	0.23504	6.2386	0.0125	1.799	1.135	2.851	theme1 high
theme1	**medium**	1	0.51125	0.25011	4.1782	0.0409	1.667	1.021	2.722	theme1 medium
percent_vacc		1	-0.58160	0.53477	1.1828	0.2768	0.559	0.196	1.594	Percent of populatin +5 vaccinated

TABLE 3: During vaccine â€" theme2

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116787

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme2	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2021	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
116787	291	4307	112189

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.844	6120.422
AIC	6306.844	6152.422
SBC	6306.844	6211.195

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	207.7734	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	104.6516	<.0001
current_gender2	1	0.6725	0.4122
raceeth	3	11.6844	0.0085
USborn	1	5.3713	0.0205
rucaccat2010new	1	2.8307	0.0925
categ	3	13.7660	0.0032
SuppressedVL2021	1	22.1802	<.0001
theme2	2	1.5710	0.4559
percent_vacc	1	0.9371	0.3330

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.57033	0.35755	2.5443	0.1107	1.769	0.878	3.565	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.43606	0.32510	19.5130	<.0001	4.204	2.223	7.951	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.21138	0.32991	44.9309	<.0001	9.128	4.782	17.426	age_hivdxrange 65 and over
current_gender2	F	1	-0.12987	0.15836	0.6725	0.4122	0.878	0.644	1.198	current_gender2 F
raceeth	Hispanic	1	0.07202	0.21428	0.1130	0.7368	1.075	0.706	1.636	raceeth Hispanic
raceeth	NHB	1	0.53052	0.17706	8.9781	0.0027	1.700	1.201	2.405	raceeth NHB
raceeth	Other	1	0.01261	0.51624	0.0006	0.9805	1.013	0.368	2.785	raceeth Other
USborn	No	1	-0.38220	0.16491	5.3713	0.0205	0.682	0.494	0.943	USborn No
rucaccat2010new	rural	1	-0.85210	0.50645	2.8307	0.0925	0.427	0.158	1.151	rucaccat2010new

										rural
categ	**IDU**	1	0.20916	0.20817	1.0096	0.3150	1.233	0.820	1.854	categ IDU
categ	**hetero**	1	0.43104	0.18426	5.4725	0.0193	1.539	1.072	2.208	categ hetero
categ	**other**	1	-0.52030	0.33772	2.3736	0.1234	0.594	0.307	1.152	categ other
SuppressedVL2021	**0**	1	0.59097	0.12548	22.1802	<.0001	1.806	1.412	2.309	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2021 specified through March 31 of the following year 0
theme2	**high**	1	0.17139	0.16639	1.0610	0.3030	1.187	0.857	1.645	theme2 high
theme2	**medium**	1	0.03069	0.18246	0.0283	0.8664	1.031	0.721	1.474	theme2 medium
percent_vacc		1	-0.52584	0.54320	0.9371	0.3330	0.591	0.204	1.714	Percent of populatin +5 vaccinated

TABLE 3: During vaccine â€" theme3

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116787

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme3	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2021	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
116787	291	4307	112189

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.844	6117.534
AIC	6306.844	6149.534
SBC	6306.844	6208.307

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	214.8823	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	105.3806	<.0001
current_gender2	1	0.5991	0.4389
raceeth	3	11.2730	0.0103
USborn	1	6.0902	0.0136
rucaccat2010new	1	3.0392	0.0813
categ	3	13.6465	0.0034
SuppressedVL2021	1	21.7198	<.0001
theme3	2	4.3326	0.1146
percent_vacc	1	3.0223	0.0821

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.56907	0.35758	2.5327	0.1115	1.767	0.877	3.561	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.43593	0.32503	19.5174	<.0001	4.204	2.223	7.948	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.21548	0.32953	45.2020	<.0001	9.166	4.805	17.485	age_hivdxrange 65 and over
current_gender2	F	1	-0.12208	0.15772	0.5991	0.4389	0.885	0.650	1.206	current_gender2 F
raceeth	Hispanic	1	0.03638	0.21688	0.0281	0.8668	1.037	0.678	1.586	raceeth Hispanic
raceeth	NHB	1	0.51005	0.18018	8.0129	0.0046	1.665	1.170	2.371	raceeth NHB
raceeth	Other	1	0.00725	0.51819	0.0002	0.9888	1.007	0.365	2.781	raceeth Other
USborn	No	1	-0.40575	0.16442	6.0902	0.0136	0.666	0.483	0.920	USborn No
rucaccat2010new	rural	1	-0.88627	0.50838	3.0392	0.0813	0.412	0.152	1.116	rucaccat2010new

										rural
categ	**IDU**	1	0.21036	0.20825	1.0203	0.3124	1.234	0.821	1.856	categ IDU
categ	**hetero**	1	0.42158	0.18497	5.1947	0.0227	1.524	1.061	2.190	categ hetero
categ	**other**	1	-0.53489	0.33814	2.5022	0.1137	0.586	0.302	1.136	categ other
SuppressedVL2021	**0**	1	0.58426	0.12536	21.7198	<.0001	1.794	1.403	2.293	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2021 specified through March 31 of the following year 0
theme3	**high**	1	0.24201	0.23295	1.0793	0.2989	1.274	0.807	2.011	theme3 high
theme3	**medium**	1	-0.09336	0.23235	0.1615	0.6878	0.911	0.578	1.436	theme3 medium
percent_vacc		1	-1.06992	0.61544	3.0223	0.0821	0.343	0.103	1.146	Percent of populatin +5 vaccinated

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116787

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme4	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2021	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
116787	291	4307	112189

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.844	6118.742
AIC	6306.844	6150.742
SBC	6306.844	6209.515

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	211.2036	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	105.1306	<.0001
current_gender2	1	0.6140	0.4333
raceeth	3	10.3577	0.0158
USborn	1	5.2278	0.0222
rucaccat2010new	1	3.3315	0.0680
categ	3	13.9135	0.0030
SuppressedVL2021	1	21.5066	<.0001
theme4	2	3.1769	0.2042
percent_vacc	1	1.5643	0.2110

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.56517	0.35752	2.4989	0.1139	1.760	0.873	3.546	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.43093	0.32490	19.3973	<.0001	4.183	2.213	7.907	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.21230	0.32941	45.1037	<.0001	9.137	4.791	17.426	age_hivdxrange 65 and over
current_gender2	F	1	-0.12344	0.15754	0.6140	0.4333	0.884	0.649	1.204	current_gender2 F
raceeth	Hispanic	1	0.05719	0.21462	0.0710	0.7899	1.059	0.695	1.613	raceeth Hispanic
raceeth	NHB	1	0.49584	0.17693	7.8538	0.0051	1.642	1.161	2.322	raceeth NHB
raceeth	Other	1	0.00427	0.51685	0.0001	0.9934	1.004	0.365	2.766	raceeth Other
USborn	No	1	-0.37865	0.16560	5.2278	0.0222	0.685	0.495	0.947	USborn No
rucaccat2010new	rural	1	-0.93583	0.51272	3.3315	0.0680	0.392	0.144	1.072	rucaccat2010new

										rural
categ	**IDU**	1	0.21131	0.20839	1.0282	0.3106	1.235	0.821	1.858	categ IDU
categ	**hetero**	1	0.44318	0.18555	5.7047	0.0169	1.558	1.083	2.241	categ hetero
categ	**other**	1	-0.50698	0.33806	2.2491	0.1337	0.602	0.311	1.168	categ other
SuppressedVL2021	**0**	1	0.58154	0.12540	21.5066	<.0001	1.789	1.399	2.287	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2021 specified through March 31 of the following year 0
theme4	**high**	1	0.40489	0.25937	2.4370	0.1185	1.499	0.902	2.492	theme4 high
theme4	**medium**	1	0.24143	0.27287	0.7829	0.3763	1.273	0.746	2.173	theme4 medium
percent_vacc		1	-0.68685	0.54917	1.5643	0.2110	0.503	0.171	1.476	Percent of populatin +5 vaccinated

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119316

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
119316	350	6743	112223

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.664	7324.756
AIC	7545.664	7330.756
SBC	7545.664	7342.330

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	196.2017	3	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	196.2017	<.0001

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.64287	0.39648	2.6291	0.1049	1.902	0.874	4.137	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.65444	0.36367	20.6955	<.0001	5.230	2.564	10.668	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.77740	0.36257	58.6807	<.0001	16.077	7.899	32.721	age_hivdxrange 65 and over

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119320

Class Level Information		
Class	Value	Design Variables
current_gender2	F	1
	M	0

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
119320	350	6745	112225

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.686	7542.462
AIC	7545.686	7544.462
SBC	7545.686	7548.320

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	3.3205	1	0.0684

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
current_gender2	1	3.3205	0.0684

Analysis of Maximum Likelihood Estimates									
								95% Hazard Ratio	

Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Confidence Limits		Label
current_gender2	F	1	0.21834	0.11982	3.3205	0.0684	1.244	0.984	1.573	current_gender2 F

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119320

Class Level Information				
Class	Value	Design Variables		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
119320	350	6745	112225

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.686	7521.992
AIC	7545.686	7527.992
SBC	7545.686	7539.566

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	22.0525	3	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
raceeth	3	22.0525	<.0001

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
raceeth	Hispanic	1	0.39459	0.17294	5.2059	0.0225	1.484	1.057	2.082	raceeth Hispanic
raceeth	NHB	1	0.68852	0.15073	20.8669	<.0001	1.991	1.482	2.675	raceeth NHB
raceeth	Other	1	0.29883	0.42887	0.4855	0.4859	1.348	0.582	3.125	raceeth Other

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119320

Class Level Information		
Class	Value	Design Variables
USborn	No	1
	Yes	0

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
119320	350	6745	112225

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.686	7542.944
AIC	7545.686	7544.944
SBC	7545.686	7548.802

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	2.6421	1	0.1041

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
USborn	1	2.6421	0.1041

Analysis of Maximum Likelihood Estimates									
			Parameter	Standard			Hazard	95% Hazard Ratio Confidence	

Parameter		DF	Estimate	Error	Chi-Square	Pr > ChiSq	Ratio	Limits		Label
USborn	No	1	-0.21123	0.12995	2.6421	0.1041	0.810	0.628	1.044	USborn No

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119320

Class Level Information				
Class	Value	Design Variables		
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
119320	350	6745	112225

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.686	7501.862
AIC	7545.686	7507.862
SBC	7545.686	7519.436

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	42.4795	3	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
categ	3	42.4795	<.0001

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
categ	IDU	1	0.84154	0.17212	23.9044	<.0001	2.320	1.656	3.251	categ IDU
categ	hetero	1	0.72335	0.12600	32.9578	<.0001	2.061	1.610	2.639	categ hetero
categ	other	1	0.11538	0.26083	0.1957	0.6582	1.122	0.673	1.871	categ other

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119320

Class Level Information		
Class	Value	Design Variables
SuppressedVL2020	0	1
	1	0

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
119320	350	6745	112225

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.686	7545.668
AIC	7545.686	7547.668
SBC	7545.686	7551.526

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	0.0172	1	0.8956

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
SuppressedVL2020	1	0.0172	0.8956

Analysis of Maximum Likelihood Estimates									
								95% Hazard	

Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Ratio Confidence Limits		Label
SuppressedVL2020	0	1	0.01549	0.11810	0.0172	0.8956	1.016	0.806	1.280	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2020 specified through March 31 of the following year 0

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119320

Class Level Information		
Class	Value	Design Variables
rucaccat2010new	rural	1
	urban	0

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
119320	350	6745	112225

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.686	7545.670
AIC	7545.686	7547.670
SBC	7545.686	7551.528

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	0.0161	1	0.8990

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
rucaccat2010new	1	0.0161	0.8990

Analysis of Maximum Likelihood Estimates									
								95% Hazard Ratio	

Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Confidence Limits		Label
rucaccat2010new	rural	1	0.04288	0.33778	0.0161	0.8990	1.044	0.538	2.024	rucaccat2010new rural

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119320

Class Level Information			
Class	Value	Design Variables	
theme	high	1	0
	low	0	0
	medium	0	1

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
119320	350	6745	112225

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7545.686	7533.293
AIC	7545.686	7537.293
SBC	7545.686	7545.008

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	11.3419	2	0.0034

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
theme	2	11.3419	0.0034

Analysis of Maximum Likelihood Estimates									

Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
theme	high	1	0.58368	0.23234	6.3113	0.0120	1.793	1.137	2.827	theme high
theme	medium	1	0.21098	0.26041	0.6564	0.4178	1.235	0.741	2.057	theme medium

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116787

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
116787	291	4307	112189

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.844	6197.342
AIC	6306.844	6203.342
SBC	6306.844	6214.362

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	99.5619	3	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	99.5619	<.0001

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.57347	0.35736	2.5752	0.1086	1.774	0.881	3.575	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.37312	0.32836	17.4874	<.0001	3.948	2.074	7.513	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.16682	0.33169	42.6759	<.0001	8.730	4.557	16.725	age_hivdxrange 65 and over

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116790

Class Level Information		
Class	Value	Design Variables
current_gender2	F	1
	M	0

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
116790	291	4308	112191

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.859	6302.369
AIC	6306.859	6304.369
SBC	6306.859	6308.042

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	4.6564	1	0.0309

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
current_gender2	1	4.6564	0.0309

Analysis of Maximum Likelihood Estimates									
								95% Hazard Ratio	

Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Confidence Limits		Label
current_gender2	F	1	0.27957	0.12956	4.6564	0.0309	1.323	1.026	1.705	current_gender2 F

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116790

Class Level Information				
Class	Value	Design Variables		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
116790	291	4308	112191

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.859	6275.966
AIC	6306.859	6281.966
SBC	6306.859	6292.986

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	29.3589	3	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
raceeth	3	29.3589	<.0001

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
raceeth	Hispanic	1	-0.27190	0.19906	1.8657	0.1720	0.762	0.516	1.126	raceeth Hispanic
raceeth	NHB	1	0.53521	0.14992	12.7444	0.0004	1.708	1.273	2.291	raceeth NHB
raceeth	Other	1	-0.16021	0.51600	0.0964	0.7562	0.852	0.310	2.342	raceeth Other

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116790

Class Level Information		
Class	Value	Design Variables
USborn	No	1
	Yes	0

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
116790	291	4308	112191

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.859	6302.697
AIC	6306.859	6304.697
SBC	6306.859	6308.370

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	3.9551	1	0.0467

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
USborn	1	3.9551	0.0467

Analysis of Maximum Likelihood Estimates									
			Parameter	Standard			Hazard	95% Hazard Ratio Confidence	

Parameter		DF	Estimate	Error	Chi-Square	Pr > ChiSq	Ratio	Limits		Label
USborn	No	1	-0.28745	0.14454	3.9551	0.0467	0.750	0.565	0.996	USborn No

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116790

Class Level Information				
Class	Value	Design Variables		
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
116790	291	4308	112191

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.859	6268.640
AIC	6306.859	6274.640
SBC	6306.859	6285.660

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	36.9155	3	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
categ	3	36.9155	<.0001

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
categ	IDU	1	0.68445	0.19849	11.8901	0.0006	1.983	1.344	2.926	categ IDU
categ	hetero	1	0.76389	0.13556	31.7530	<.0001	2.147	1.646	2.800	categ hetero
categ	other	1	-0.07053	0.30680	0.0528	0.8182	0.932	0.511	1.700	categ other

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116790

Class Level Information		
Class	Value	Design Variables
SuppressedVL2021	0	1
	1	0

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
116790	291	4308	112191

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.859	6289.357
AIC	6306.859	6291.357
SBC	6306.859	6295.031

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	18.1145	1	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
SuppressedVL2021	1	18.1145	<.0001

Analysis of Maximum Likelihood Estimates									
								95% Hazard	

Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Ratio Confidence Limits		Label
SuppressedVL2021	0	1	0.52056	0.12231	18.1145	<.0001	1.683	1.324	2.139	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2021 specified through March 31 of the following year 0

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116790

Class Level Information		
Class	Value	Design Variables
rucaccat2010new	rural	1
	urban	0

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
116790	291	4308	112191

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.859	6305.103
AIC	6306.859	6307.103
SBC	6306.859	6310.777

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	1.4324	1	0.2314

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
rucaccat2010new	1	1.4324	0.2314

Analysis of Maximum Likelihood Estimates									
								95% Hazard Ratio	

Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Confidence Limits		Label
rucaccat2010new	rural	1	-0.60273	0.50361	1.4324	0.2314	0.547	0.204	1.469	rucaccat2010new rural

TABLE 3: During vaccine â€" theme4

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_MARCH	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116791
Number of Observations Used	116790

Class Level Information			
Class	Value	Design Variables	
theme	high	1	0
	low	0	0
	medium	0	1

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
116790	291	4308	112191

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6306.859	6296.267
AIC	6306.859	6300.267
SBC	6306.859	6307.613

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	9.3612	2	0.0093

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
theme	2	9.3612	0.0093

Analysis of Maximum Likelihood Estimates									

Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
theme	high	1	0.68929	0.26730	6.6494	0.0099	1.992	1.180	3.364	theme high
theme	medium	1	0.38503	0.29431	1.7115	0.1908	1.470	0.825	2.617	theme medium

Supporting Table 1b: Univariate crude sub-distribution HRs

ClassVal0	ProbChiSq	HazardRatio	HRLowerCL	HRUpperCL	period	varname
35-49	0.1049	1.902	0.874	4.137	Before	age_hivdxrange
50-64	<.0001	5.230	2.564	10.668	Before	age_hivdxrange
65 and over	<.0001	16.077	7.899	32.721	Before	age_hivdxrange
F	0.0684	1.244	0.984	1.573	Before	current_gender2
Hispanic	0.0225	1.484	1.057	2.082	Before	raceeth
NHB	<.0001	1.991	1.482	2.675	Before	raceeth
Other	0.4859	1.348	0.582	3.125	Before	raceeth
No	0.1041	0.810	0.628	1.044	Before	USborn
IDU	<.0001	2.320	1.656	3.251	Before	categ
hetero	<.0001	2.061	1.610	2.639	Before	categ
other	0.6582	1.122	0.673	1.871	Before	categ
0	0.8956	1.016	0.806	1.280	Before	SuppressedVL2020
rural	0.8990	1.044	0.538	2.024	Before	rucaccat2010new
high	0.0120	1.793	1.137	2.827	Before	theme
medium	0.4178	1.235	0.741	2.057	Before	theme
35-49	0.1086	1.774	0.881	3.575	During	age_hivdxrange
50-64	<.0001	3.948	2.074	7.513	During	age_hivdxrange
65 and over	<.0001	8.730	4.557	16.725	During	age_hivdxrange
F	0.0309	1.323	1.026	1.705	During	current_gender2
Hispanic	0.1720	0.762	0.516	1.126	During	raceeth
NHB	0.0004	1.708	1.273	2.291	During	raceeth
Other	0.7562	0.852	0.310	2.342	During	raceeth
No	0.0467	0.750	0.565	0.996	During	USborn
IDU	0.0006	1.983	1.344	2.926	During	categ
hetero	<.0001	2.147	1.646	2.800	During	categ
other	0.8182	0.932	0.511	1.700	During	categ
0	<.0001	1.683	1.324	2.139	During	SuppressedVL2021
rural	0.2314	0.547	0.204	1.469	During	rucaccat2010new
high	0.0099	1.992	1.180	3.364	During	theme
medium	0.1908	1.470	0.825	2.617	During	theme

4-WK During (expect COVID=304)

The FREQ Procedure

censor1	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	115408	95.95	115408	95.95
1	304	0.25	115712	96.20
2	4566	3.80	120278	100.00

TABLE 4: Before vaccine Individual Model (4-week sensitivity)

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_4WK	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119316

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
SuppressedVL2020	0	1		
	1	0		

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
119316	340	6753	112223

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7313.297	7031.376
AIC	7313.297	7057.376
SBC	7313.297	7107.152

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	286.6483	13	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	194.6035	<.0001
current_gender2	1	2.2856	0.1306
raceeth	3	23.1392	<.0001
USborn	1	11.5779	0.0007
rucaccat2010new	1	0.0263	0.8711
categ	3	12.1622	0.0068
SuppressedVL2020	1	0.4588	0.4982

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.57496	0.39847	2.0821	0.1490	1.777	0.814	3.880	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.65710	0.36182	20.9760	<.0001	5.244	2.580	10.657	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.81347	0.36272	60.1633	<.0001	16.668	8.187	33.933	age_hivdxrange 65 and over
current_gender2	F	1	-0.21354	0.14125	2.2856	0.1306	0.808	0.612	1.065	current_gender2 F
raceeth	Hispanic	1	0.83034	0.19010	19.0781	<.0001	2.294	1.581	3.330	raceeth Hispanic
raceeth	NHB	1	0.79202	0.17797	19.8043	<.0001	2.208	1.558	3.129	raceeth NHB
raceeth	Other	1	0.62130	0.43213	2.0672	0.1505	1.861	0.798	4.342	raceeth Other
USborn	No	1	-0.48496	0.14252	11.5779	0.0007	0.616	0.466	0.814	USborn No
rucaccat2010new	rural	1	-0.05493	0.33850	0.0263	0.8711	0.947	0.488	1.838	rucaccat2010new rural
categ	IDU	1	0.37701	0.19064	3.9108	0.0480	1.458	1.003	2.118	categ IDU
categ	hetero	1	0.44347	0.16823	6.9485	0.0084	1.558	1.120	2.167	categ hetero
categ	other	1	-0.18455	0.29014	0.4046	0.5247	0.831	0.471	1.468	categ other
SuppressedVL2020	0	1	0.08417	0.12427	0.4588	0.4982	1.088	0.853	1.388	PWH with a

										suppressed VL (<200 copies/mL) on the last VL from January 1 of 2020 specified through March 31 of the following year 0

TABLE 4: Before vaccine â€" Final Model (4-week sensitivity)

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_4WK	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119316

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2020	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
119316	340	6753	112223

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7313.297	7026.491
AIC	7313.297	7058.491
SBC	7313.297	7119.755

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	293.4164	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	193.9610	<.0001
current_gender2	1	2.4106	0.1205
raceeth	3	20.3495	0.0001
USborn	1	11.1925	0.0008
rucaccat2010new	1	0.1376	0.7107
categ	3	11.6552	0.0087
SuppressedVL2020	1	0.4190	0.5174
theme	2	4.3239	0.1151
percent_vacc	1	0.2948	0.5872

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.57060	0.39848	2.0504	0.1522	1.769	0.810	3.864	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.64940	0.36189	20.7725	<.0001	5.204	2.560	10.577	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.80681	0.36268	59.8926	<.0001	16.557	8.133	33.705	age_hivdxrange 65 and over
current_gender2	F	1	-0.21901	0.14106	2.4106	0.1205	0.803	0.609	1.059	current_gender2 F
raceeth	Hispanic	1	0.80888	0.19120	17.8972	<.0001	2.245	1.544	3.266	raceeth Hispanic
raceeth	NHB	1	0.72844	0.18107	16.1846	<.0001	2.072	1.453	2.954	raceeth NHB
raceeth	Other	1	0.59305	0.43085	1.8946	0.1687	1.809	0.778	4.210	raceeth Other
USborn	No	1	-0.49210	0.14709	11.1925	0.0008	0.611	0.458	0.816	USborn No
rucaccat2010new	rural	1	-0.12939	0.34879	0.1376	0.7107	0.879	0.444	1.741	rucaccat2010new

										rural
categ	**IDU**	1	0.37029	0.19046	3.7799	0.0519	1.448	0.997	2.103	categ IDU
categ	**hetero**	1	0.43437	0.16860	6.6372	0.0100	1.544	1.110	2.149	categ hetero
categ	**other**	1	-0.17874	0.29093	0.3775	0.5390	0.836	0.473	1.479	categ other
SuppressedVL2020	**0**	1	0.08078	0.12479	0.4190	0.5174	1.084	0.849	1.385	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2020 specified through March 31 of the following year 0
theme	**high**	1	0.37466	0.24053	2.4262	0.1193	1.454	0.908	2.331	theme high
theme	**medium**	1	0.09825	0.26052	0.1422	0.7061	1.103	0.662	1.838	theme medium
percent_vacc		1	-0.29707	0.54716	0.2948	0.5872	0.743	0.254	2.171	Percent of populatin +5 vaccinated

TABLE 4: During vaccine â€" Individual Model (4-week sensitivity)

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_4WK	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116967
Number of Observations Used	116963

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
SuppressedVL2021	0	1		
	1	0		

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
116963	301	4471	112191

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6539.323	6353.583
AIC	6539.323	6379.583
SBC	6539.323	6427.775

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	206.0356	13	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	105.2543	<.0001
current_gender2	1	0.1536	0.6951
raceeth	3	12.0319	0.0073
USborn	1	7.3830	0.0066
rucaccat2010new	1	2.5485	0.1104
categ	3	13.7921	0.0032
SuppressedVL2021	1	20.8225	<.0001

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.61774	0.35538	3.0215	0.0822	1.855	0.924	3.722	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.47593	0.32439	20.7010	<.0001	4.375	2.317	8.262	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.22794	0.32930	45.7733	<.0001	9.281	4.867	17.697	age_hivdxrange 65 and over
current_gender2	F	1	-0.06070	0.15488	0.1536	0.6951	0.941	0.695	1.275	current_gender2 F
raceeth	Hispanic	1	0.07644	0.20919	0.1335	0.7148	1.079	0.716	1.627	raceeth Hispanic
raceeth	NHB	1	0.52661	0.17396	9.1639	0.0025	1.693	1.204	2.381	raceeth NHB
raceeth	Other	1	-0.02015	0.51604	0.0015	0.9689	0.980	0.356	2.695	raceeth Other
USborn	No	1	-0.42302	0.15568	7.3830	0.0066	0.655	0.483	0.889	USborn No
rucaccat2010new	rural	1	-0.80738	0.50575	2.5485	0.1104	0.446	0.166	1.202	rucaccat2010new rural
categ	IDU	1	0.19894	0.20541	0.9380	0.3328	1.220	0.816	1.825	categ IDU
categ	hetero	1	0.41230	0.18265	5.0953	0.0240	1.510	1.056	2.160	categ hetero
categ	other	1	-0.55561	0.33583	2.7372	0.0980	0.574	0.297	1.108	categ other
SuppressedVL2021	0	1	0.56499	0.12382	20.8225	<.0001	1.759	1.380	2.243	PWH with a

										suppressed VL (<200 copies/mL) on the last VL from January 1 of 2021 specified through March 31 of the following year 0

TABLE 4: During vaccine â€" Final Model (4-week sensitivity)

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_4WK	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116967
Number of Observations Used	116963

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2021	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
116963	301	4471	112191

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6539.323	6347.584
AIC	6539.323	6379.584
SBC	6539.323	6438.898

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	214.5025	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	105.9142	<.0001
current_gender2	1	0.1929	0.6605
raceeth	3	9.5636	0.0227
USborn	1	6.2669	0.0123
rucaccat2010new	1	3.2612	0.0709
categ	3	13.2843	0.0041
SuppressedVL2021	1	20.9453	<.0001
theme	2	5.2826	0.0713
percent_vacc	1	1.0249	0.3114

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.61582	0.35544	3.0017	0.0832	1.851	0.922	3.715	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.47619	0.32449	20.6952	<.0001	4.376	2.317	8.266	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.23193	0.32917	45.9746	<.0001	9.318	4.888	17.763	age_hivdxrange 65 and over
current_gender2	F	1	-0.06806	0.15498	0.1929	0.6605	0.934	0.689	1.266	current_gender2 F
raceeth	Hispanic	1	0.06203	0.20846	0.0885	0.7660	1.064	0.707	1.601	raceeth Hispanic
raceeth	NHB	1	0.46918	0.17436	7.2409	0.0071	1.599	1.136	2.250	raceeth NHB
raceeth	Other	1	-0.04742	0.51561	0.0085	0.9267	0.954	0.347	2.620	raceeth Other
USborn	No	1	-0.40738	0.16273	6.2669	0.0123	0.665	0.484	0.915	USborn No
rucaccat2010new	rural	1	-0.92440	0.51188	3.2612	0.0709	0.397	0.145	1.082	rucaccat2010new

										rural
categ	**IDU**	1	0.18160	0.20537	0.7819	0.3766	1.199	0.802	1.793	categ IDU
categ	**hetero**	1	0.39737	0.18217	4.7581	0.0292	1.488	1.041	2.126	categ hetero
categ	**other**	1	-0.55711	0.33703	2.7325	0.0983	0.573	0.296	1.109	categ other
SuppressedVL2021	**0**	1	0.56420	0.12328	20.9453	<.0001	1.758	1.381	2.239	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2021 specified through March 31 of the following year 0
theme	**high**	1	0.58542	0.26834	4.7596	0.0291	1.796	1.061	3.038	theme high
theme	**medium**	1	0.41434	0.29122	2.0243	0.1548	1.513	0.855	2.678	theme medium
percent_vacc		1	-0.55855	0.55172	1.0249	0.3114	0.572	0.194	1.687	Percent of populatin +5 vaccinated

8-WK During (expect COVID=284)

The FREQ Procedure

censor1	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	115398	96.24	115398	96.24
1	284	0.24	115682	96.48
2	4221	3.52	119903	100.00

TABLE 5: Before vaccine â€" Individual Model (8-week sensitivity)

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_8WK	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119316

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
SuppressedVL2020	0	1		
	1	0		

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
119316	360	6733	112223

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7778.029	7483.993
AIC	7778.029	7509.993
SBC	7778.029	7560.512

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	296.0893	13	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	202.9477	<.0001
current_gender2	1	1.2496	0.2636
raceeth	3	24.5384	<.0001
USborn	1	11.9967	0.0005
rucaccat2010new	1	0.1102	0.7399
categ	3	12.1910	0.0068
SuppressedVL2020	1	0.4810	0.4880

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.63976	0.39569	2.6141	0.1059	1.896	0.873	4.118	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.74091	0.36119	23.2313	<.0001	5.703	2.809	11.575	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.86627	0.36209	62.6614	<.0001	17.571	8.642	35.729	age_hivdxrange 65 and over
current_gender2	F	1	-0.15391	0.13769	1.2496	0.2636	0.857	0.655	1.123	current_gender2 F
raceeth	Hispanic	1	0.82448	0.18426	20.0224	<.0001	2.281	1.589	3.273	raceeth Hispanic
raceeth	NHB	1	0.79002	0.17155	21.2075	<.0001	2.203	1.574	3.084	raceeth NHB
raceeth	Other	1	0.55266	0.43054	1.6478	0.1993	1.738	0.747	4.041	raceeth Other
USborn	No	1	-0.47798	0.13800	11.9967	0.0005	0.620	0.473	0.813	USborn No
rucaccat2010new	rural	1	-0.11227	0.33818	0.1102	0.7399	0.894	0.461	1.734	rucaccat2010new rural
categ	IDU	1	0.33034	0.18470	3.1988	0.0737	1.391	0.969	1.998	categ IDU
categ	hetero	1	0.40244	0.16361	6.0503	0.0139	1.495	1.085	2.061	categ hetero
categ	other	1	-0.27393	0.28780	0.9060	0.3412	0.760	0.433	1.337	categ other
SuppressedVL2020	0	1	0.08350	0.12039	0.4810	0.4880	1.087	0.859	1.376	PWH with a

										suppressed VL (<200 copies/mL) on the last VL from January 1 of 2020 specified through March 31 of the following year 0

TABLE 5: Before vaccine â€" Final Model (8-week sensitivity)

The PHREG Procedure

Model Information		
Data Set	WORK.BEFORE_GROUP_8WK	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	119321
Number of Observations Used	119316

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2020	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
119316	360	6733	112223

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7778.029	7479.237
AIC	7778.029	7511.237
SBC	7778.029	7573.415

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	300.7634	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	201.5752	<.0001
current_gender2	1	1.3517	0.2450
raceeth	3	21.2948	<.0001
USborn	1	12.1548	0.0005
rucaccat2010new	1	0.1848	0.6673
categ	3	11.8275	0.0080
SuppressedVL2020	1	0.3947	0.5298
theme	2	4.1510	0.1255
percent_vacc	1	0.0166	0.8975

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.63445	0.39573	2.5704	0.1089	1.886	0.868	4.096	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.73019	0.36121	22.9444	<.0001	5.642	2.779	11.452	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.85427	0.36201	62.1659	<.0001	17.362	8.540	35.297	age_hivdxrange 65 and over
current_gender2	F	1	-0.15981	0.13746	1.3517	0.2450	0.852	0.651	1.116	current_gender2 F
raceeth	Hispanic	1	0.79180	0.18472	18.3732	<.0001	2.207	1.537	3.170	raceeth Hispanic
raceeth	NHB	1	0.72295	0.17356	17.3511	<.0001	2.060	1.466	2.895	raceeth NHB
raceeth	Other	1	0.52218	0.42909	1.4809	0.2236	1.686	0.727	3.909	raceeth Other
USborn	No	1	-0.49975	0.14334	12.1548	0.0005	0.607	0.458	0.803	USborn No
rucaccat2010new	rural	1	-0.14915	0.34698	0.1848	0.6673	0.861	0.436	1.700	rucaccat2010new

										rural
categ	**IDU**	1	0.32974	0.18453	3.1931	0.0740	1.391	0.969	1.997	categ IDU
categ	**hetero**	1	0.39837	0.16406	5.8963	0.0152	1.489	1.080	2.054	categ hetero
categ	**other**	1	-0.26284	0.28861	0.8294	0.3624	0.769	0.437	1.354	categ other
SuppressedVL2020	**0**	1	0.07615	0.12120	0.3947	0.5298	1.079	0.851	1.368	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2020 specified through March 31 of the following year 0
theme	**high**	1	0.37152	0.23359	2.5297	0.1117	1.450	0.917	2.292	theme high
theme	**medium**	1	0.12451	0.25354	0.2412	0.6234	1.133	0.689	1.862	theme medium
percent_vacc		1	-0.06847	0.53153	0.0166	0.8975	0.934	0.329	2.647	Percent of populatin +5 vaccinated

TABLE 5: During vaccine â€" Individual Model (8-week sensitivity)

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_8WK	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116599
Number of Observations Used	116595

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
SuppressedVL2021	0	1		
	1	0		

Summary of Failure Outcomes			
Total	Event of Interest	Competing Event	Censored
116595	281	4133	112181

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6074.412	5903.945
AIC	6074.412	5929.945
SBC	6074.412	5977.243

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	194.1876	13	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	95.8605	<.0001
current_gender2	1	0.6665	0.4143
raceeth	3	11.2592	0.0104
USborn	1	6.7769	0.0092
rucaccat2010new	1	2.1508	0.1425
categ	3	13.5163	0.0036
SuppressedVL2021	1	18.3297	<.0001

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.56343	0.35770	2.4811	0.1152	1.757	0.871	3.541	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.38118	0.32505	18.0555	<.0001	3.980	2.105	7.525	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.14326	0.33041	42.0778	<.0001	8.527	4.462	16.295	age_hivdxrange 65 and over
current_gender2	F	1	-0.13025	0.15955	0.6665	0.4143	0.878	0.642	1.200	current_gender2 F
raceeth	Hispanic	1	0.01758	0.21870	0.0065	0.9359	1.018	0.663	1.562	raceeth Hispanic
raceeth	NHB	1	0.51359	0.18082	8.0674	0.0045	1.671	1.173	2.382	raceeth NHB
raceeth	Other	1	0.03931	0.51742	0.0058	0.9394	1.040	0.377	2.867	raceeth Other
USborn	No	1	-0.42193	0.16208	6.7769	0.0092	0.656	0.477	0.901	USborn No
rucaccat2010new	rural	1	-0.74242	0.50623	2.1508	0.1425	0.476	0.176	1.284	rucaccat2010new rural
categ	IDU	1	0.25742	0.21322	1.4575	0.2273	1.294	0.852	1.965	categ IDU
categ	hetero	1	0.46891	0.18864	6.1790	0.0129	1.598	1.104	2.313	categ hetero
categ	other	1	-0.43934	0.33854	1.6842	0.1944	0.644	0.332	1.251	categ other
SuppressedVL2021	0	1	0.55159	0.12884	18.3297	<.0001	1.736	1.349	2.235	PWH with a

										suppressed VL (<200 copies/mL) on the last VL from January 1 of 2021 specified through March 31 of the following year 0

TABLE 5: During vaccine â€" Final Model (8-week sensitivity)

The PHREG Procedure

Model Information		
Data Set	WORK.AFTER_GROUP_8WK	
Dependent Variable	time_diff	
Status Variable	censor1	
Event of Interest	1	
Competing Event	2	
Censored Value	0	
Weight Variable	mi_weight	Multiple imputation weight

Number of Observations Read	116599
Number of Observations Used	116595

Class Level Information				
Class	Value	Design Variables		
age_hivdxrange	18-34	0	0	0
	35-49	1	0	0
	50-64	0	1	0
	65 and over	0	0	1
current_gender2	F	1		
	M	0		
raceeth	Hispanic	1	0	0
	NHB	0	1	0
	NHW	0	0	0
	Other	0	0	1
categ	IDU	1	0	0
	MSM	0	0	0
	hetero	0	1	0
	other	0	0	1
USborn	No	1		
	Yes	0		
rucaccat2010new	rural	1		
	urban	0		
theme	high	1	0	
	low	0	0	
	medium	0	1	
SuppressedVL2021	0	1		
	1	0		

Summary of Failure Outcomes			
	Event of	Competing	

Total	Interest	Event	Censored
116595	281	4133	112181

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	6074.412	5897.016
AIC	6074.412	5929.016
SBC	6074.412	5987.230

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Wald	205.2634	16	<.0001

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_hivdxrange	3	97.0317	<.0001
current_gender2	1	0.7387	0.3901
raceeth	3	8.8759	0.0310
USborn	1	5.2954	0.0214
rucaccat2010new	1	3.1427	0.0763
categ	3	12.8731	0.0049
SuppressedVL2021	1	18.8381	<.0001
theme	2	5.3814	0.0678
percent_vacc	1	2.2842	0.1307

Analysis of Maximum Likelihood Estimates										
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
age_hivdxrange	35-49	1	0.56305	0.35776	2.4769	0.1155	1.756	0.871	3.540	age_hivdxrange 35-49
age_hivdxrange	50-64	1	1.38566	0.32530	18.1447	<.0001	3.997	2.113	7.563	age_hivdxrange 50-64
age_hivdxrange	65 and over	1	2.15485	0.33039	42.5375	<.0001	8.627	4.514	16.484	age_hivdxrange 65 and over
current_gender2	F	1	-0.13728	0.15973	0.7387	0.3901	0.872	0.637	1.192	current_gender2 F
raceeth	Hispanic	1	0.01778	0.21882	0.0066	0.9353	1.018	0.663	1.563	raceeth Hispanic
raceeth	NHB	1	0.46013	0.18224	6.3753	0.0116	1.584	1.108	2.264	raceeth NHB
raceeth	Other	1	0.01306	0.51721	0.0006	0.9798	1.013	0.368	2.792	raceeth Other
USborn	No	1	-0.38725	0.16828	5.2954	0.0214	0.679	0.488	0.944	USborn No
rucaccat2010new	rural	1	-0.91065	0.51369	3.1427	0.0763	0.402	0.147	1.101	rucaccat2010new

										rural
categ	**IDU**	1	0.23310	0.21314	1.1960	0.2741	1.263	0.831	1.917	categ IDU
categ	**hetero**	1	0.44829	0.18797	5.6879	0.0171	1.566	1.083	2.263	categ hetero
categ	**other**	1	-0.44592	0.33979	1.7222	0.1894	0.640	0.329	1.246	categ other
SuppressedVL2021	**0**	1	0.55489	0.12785	18.8381	<.0001	1.742	1.356	2.238	PWH with a suppressed VL (<200 copies/mL) on the last VL from January 1 of 2021 specified through March 31 of the following year 0
theme	**high**	1	0.60463	0.27893	4.6988	0.0302	1.831	1.060	3.162	theme high
theme	**medium**	1	0.40384	0.30183	1.7902	0.1809	1.498	0.829	2.706	theme medium
percent_vacc		1	-0.86238	0.57059	2.2842	0.1307	0.422	0.138	1.292	Percent of populatin +5 vaccinated